KUNAL SHARAD BADGUJAR

AI/ML Engineer • Applied AI Builder • Open-Source Contributor

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■ PROFESSIONAL SUMMARY

B.Tech AI & ML undergraduate who ships end-to-end products — from React and Next.js interfaces to FastAPI and Firebase backends — powered by LLM APIs, RAG pipelines, and computer vision. Open-source contributor at Vicharanashala, IIT Ropar, with **8 live, deployed projects** across education, productivity, fitness, and spatial computing.

■ EDUCATION

B.Tech — Computer Science & Engineering (AI & ML) | *MIT Academy of Engineering, Alandi, Pune* **2025 – 2029**

■ TECHNICAL SKILLS

Languages	C, C++, Python, JavaScript, TypeScript, SQL, HTML5, CSS3
Frontend	React, Next.js, Vite, Tailwind CSS, Three.js, Responsive & Interactive UI
Backend & APIs	Node.js, Express, FastAPI, REST APIs, Prisma ORM
AI / GenAI	LLM Integration (Groq, Gemini, MiniMax), RAG, Prompt Engineering, AI Agents, MediaPipe
Databases & Cloud	Firebase, Supabase, PostgreSQL, Vercel
Tools & Practices	Git, GitHub (PR Workflow), Linux, VS Code, Monaco Editor, Piston Sandboxing, DSA

■ EXPERIENCE

Vicharanashala, IIT Ropar **May 2026 – Present**

Open-Source & AI Development Intern | Remote

- Contribute to open-source learning and community platforms, including **Tenali** (a mathematics quiz platform) and **crowd-source-faq**.
- Built interactive **Bayes' Theorem** and **Normal Distribution** learning modules using the MiniMax API, and designed an animated Tenali Raman mascot system with multiple emotion states to make learning more engaging.
- Designed and submitted **Yaksha AI**, a full-stack FAQ portal concept featuring a 3D knowledge graph, a Groq-powered RAG chatbot, and participation-focused gamification.

IBM *Intern*

- Completed an IBM internship program; further details are available on LinkedIn.

OBLIQ *Open-Source Developer*

- Contributed to OBLIQ's website project through the organization's public GitHub repositories and pull-request workflow.

■ PROJECTS

Optima AI | Live Demo *Next.js · React · TypeScript · Groq · Piston*
AI-powered code optimizer that refactors code, runs it in a sandbox, verifies behavior, and benchmarks the improvement.

ARQuest | Live Demo *React · Tailwind CSS · Supabase · MediaPipe*
Gamified fitness and habit app with quests, XP, and real-time exercise feedback using camera-based pose tracking.

Saras AI | Live Demo *Next.js · Tailwind CSS · Groq · Prisma*
Active-learning platform for video-based study that combines checkpoints, quizzes, flashcards, and focused revision.

EduCore | Live Demo *React · Vite · Tailwind CSS · Firebase*
Role-based team workspace with project organization, task management, chat, collaboration, and a responsive dashboard.

Ascend | Live Demo *React · Firebase · Monaco Editor · Piston*
DSA practice environment with in-browser coding, test-case execution, submission tracking, and progress dashboards.

PivotVault | Live Demo *Gemini AI · RAG · PostgreSQL · FastAPI*
AI-assisted platform for exploring startup post-mortems and surfacing evidence-grounded risk patterns.

PathForge AI | Live Demo *React · Vite · Tailwind CSS · Three.js*
Interactive path-planning visualizer with route exploration, dashboards, and immersive spatial views.

Locentra | Live Demo *React · TypeScript · Three.js · Python · FastAPI*
Indoor-positioning research interface for anchor placement, trajectory analysis, 2D/3D views, and solver evaluation.

■ CERTIFICATIONS

IBM Build an AI Agent | AI Fluency: Framework & Foundations | Claude 101
Additional Python Essentials | Data Analytics Essentials | Introduction to Modern AI